

Alessandro Armando Vigliano

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Google Scholar
ORCID
ResearchGate profile

RESEARCH FOCUS

High-energy astrophysicist specializing in Gamma-Ray Bursts, fast transients, and multi-instrument X-ray/gamma-ray datasets. My research combines automated data-analysis pipelines, temporal/spectral characterization, and statistical methods for large transient samples from Swift, Fermi, MAGIC, and CTAO-LST1. I have previous experience in gamma-ray polarimetry in the context of MeV mission concepts, and I am interested in using spectroscopic and polarimetric observations to study relativistic jets and high-energy transient populations.

APPOINTMENTS

09/2026 – CURRENT

Postdoctoral Research Associate - Incarico di Ricerca

Research within the COSI collaboration.
INAF Osservatorio Astronomico di Brera, Merate
Project: *"COSI: simulazione ed analisi dati dell'emissione di getti relativistici"*
Responsible: Fabrizio Tavecchio

12/2024 – 09/2026

Postdoctoral Research Associate - Assegno di Ricerca

Research within the Fermi, MAGIC, and CTAO collaborations.
Faculty of Physics, University of Trieste
Project: *"Spectral analysis of Gamma-Ray Bursts and Solar Flares emitting in Gamma Rays"*
Responsible: Francesco Longo

EDUCATION

10/2021 – 10/2024

PhD in Mathematics and Physics

Faculty of Mathematical and Physical Sciences, University of Udine
Supervisor: Barbara De Lotto
Co-Supervisors: Francesco Longo, Stefano Ansoldi, Andrea Vacchi
Thesis: ***Novel strategies for the observations of fast gamma-ray transients***

2017 – 2021

Master's degree in Physics

Faculty of Physics, University of Trieste
Curriculum: Astrophysics and Cosmology
Thesis: ***Gamma-Ray Burst polarization studies with AMEGO***
Supervisor: Stefano Ansoldi
Co-Supervisor: Francesco Longo

2014 – 2017

Bachelor's degree in Physics

Faculty of Physics, University of Trieste
Thesis: ***Estimation of the contribution of AGN host galaxies to reionization.***
Original title: *Stima del contributo di galassie ospitanti AGN alla reionizzazione.*
Supervisor: Pierluigi Monaco
Co-Supervisor: Fabio Fontanot

INTERNSHIPS

06/2020 – 09/2020

MEGAlib for simulation of polarized MeV photons for gamma-ray polarimetry studies

Master Internship
INFN, section of Trieste

02/07/2018 – 07/07/2018

Internship at Osservatorio Astrofisico e Astronomico di Asiago

Master Internship
Trieste University

01/09/2016 – 15/02/2017

Satellite constellation system designing

Bachelor Internship
PicoSaTs
https://picosats.eu/

SCHOOLS

08/10/2023 – 14/10/2023

5th MAGIC Hardware School 2023

MAGIC collaboration, La Palma, Canary Islands

12/02/2023 – 21/02/2023

Winter School of Theoretical Physics and third COST Action CA18108 Training School ”Gravity – Classical, Quantum and Phenomenology”

COST Action CA 18108 QGMM, University of Wroclaw, Poland

03/09/2022 – 10/09/2022

Quantum gravity phenomenology in the multi-messenger approach - COST CA18108 Second Training School

COST Action CA18108 , Faculty of Physics, Belgrade (Serbia)

28/03/2022 – 08/04/2022

Astrophysical sources of cosmic rays - ISAPP School Paris

Institut Pascal, Université Paris Saclay

WORKSHOPS

03/04/2023 – 07/04/2023

Workshop: Machine learning enabled searches for new physics in astrophysical data.

IFPU, Trieste

09/01/2023 – 13/01/2023

Workshop ”High redshift Gamma-Ray-Bursts in the JWST era”

Sexten Center for Astrophysics

18/07/2022 – 22/07/2022

Workshop: Hands On the Extreme Universe with High Energy Gamma Rays

Sexten Center for Astrophysics

23/05/2022 – 26/05/2022

Gravitational Wave Open data Workshop 2022

LIGO-Virgo-KAGRA, Trieste University

01/02/2022 – 03/02/2022

International workshop on VHE Data Analysis using Open Source Packages

University of Kashmir

TALKS, SEMINARS AND CONTRIBUTIONS

20/07/2026 – 24/07/2026

From 9 to 40: the SST Telescopes and Gamma-Ray Science with ASTRI and CTAO

Contributed talk: *Reassessing GRB high-energy emission correlations using a large, homogeneous sample of X-ray afterglows*
Sexten, Dolomites, Italy

26/05/2026 – 28/05/2026

Invited seminar

Invited seminar: *Correlations in Gamma-Ray Bursts*
University of Palermo & INAF Palermo, Italy

24/03/2026 – 26/03/2026

Theseus Conference 2026

Contributed talk: *Reassessing GRB high-energy emission correlations using a large, homogeneous sample of X-ray afterglows*
Malaga, Spain

19/01/2026 – 23/01/2026

Gamma-Ray Bursts: the Great Gig in the Sky

Contributed talk: *Reassessing GRB high-energy emission correlations using a large, homogeneous sample of X-ray afterglows*
Sexten, Dolomites, Italy

17/11/2025 – 21/11/2025

2nd LST+MAGIC F2F Science Meeting

Contributed Talk: *The Soccer Ball Tessellation for the follow-up of Transient Events*
Granada, Spain

30/06/2025 – 04/07/2025

Advances in Modeling High-Energy Astrophysical Sources: Insights from recent multimessenger discoveries

Contributed Talk: *Insight on GRB physics from a novel data-driven method for systematic analysis of X-ray light-curves*
Sexten, Dolomites, Italy

24/03/2025 – 28/03/2025

Celebrating 20 years of Swift Discoveries

Presented a poster: *Insight on GRB physics from a novel data driven method for systematic analysis of X-ray light-curves*
Firenze, Italy

15/01/2024 – 17/01/2024

1st VHEGAM meeting: LST proposals by the Italian community

Contributed Talk: Alternative tiling strategies for transient events
LST collaboration, Bologna, Italy

26/07/2023 – 03/08/2023

38th International Cosmic Ray Conference (ICRC2023)

Presented a poster and proceeding on *Gamma-Ray Burst localization with GECCO's BGO anticoincidence shields*
Institute for Cosmic Ray Research, University of Tokyo, Japan

19/06/2023 – 23/06/2023

ASAPP2023 - Advances in Space AstroParticle Physics. Frontier technologies for particle measurements in space. International Conference

Contributed Talk: *Gamma-Ray Burst localization with GECCO's BGO anticoincidence shields*
Agenzia Spaziale Italiana (ASI), Istituto Nazionale di Fisica Nucleare (INFN), Perugia

SELECTED RESEARCH ACTIVITIES

09/2026 – ONGOING

Multimessenger constraints on GRB jet structure

Collaborators: Fabrizio Tavecchio, Giancarlo Ghirlanda, Lara Nava

Constraints on the structure of the GRB jet through the joint use of constraints on GW emission from binary neutron stars (short GRB), temporal afterglow characteristics and prompt phase polarimetry with COSI.

05/2026 – ONGOING

Testing the origin of GRB X-ray flares with a large sample analysis

Collaborators: Francesco Longo, Željka Bošnjak

Automated spectro-temporal analysis of X-ray flaring episodes in Swift-XRT GRB afterglows, based on afterglow-model subtraction, flare-window selection, and dedicated spectral extraction. The project investigates whether flare spectra are statistically closer to prompt emission than to the underlying afterglow, providing a large-sample statistically robust test of their possible late central-engine origin.

01/2024 – 05/2026

Reassessing GRB high-energy emission correlations using a large, homogeneous sample of X-ray afterglows

Collaborators: Francesco Longo, Željka Bošnjak

Paper

Development of a data-driven and model-independent framework for the automated analysis of Swift-XRT GRB afterglows. The pipeline enables homogeneous characterization of X-ray light curves for an order of magnitude more events than previous studies, allowing statistically robust investigations of GRB temporal and spectral correlations and of the impact of observational biases on large-scale inference.

01/2024 – 10/2025

Development of the divergent mode observation for CTAO

Collaborators: Francesco Longo & INFN Divergent Pointing team

Development and optimization of divergent pointing strategies for Imaging Atmospheric Cherenkov Telescopes, with applications to transient follow-up observations and wide-field extragalactic surveys for CTAO. Comparison with tessellation-based observational strategies and evaluation of survey efficiency and sky coverage.

11/2021 – 05/2025

New tessellation strategies for multi-messenger observations of VHE fast transients with IACTs, PhD project

Collaborators: Francesco Longo

PhD Thesis

Development of the “Soccer Ball Tessellation” strategy, a novel equal-area sky tessellation framework based on Goldberg polyhedra for the follow-up of poorly localized transient events such as GRBs and gravitational-wave counterparts. The method optimizes sky coverage while minimizing the number of telescope pointings, improving observational efficiency for narrow field-of-view instruments such as CTAO and other IACTs.

05/2021 – 08/2023

GRB localization capability for GECCO

Team: A. A. Vigliano, F. Longo, I. Martinez, E. A. Yates, A. Moiseev, A. Zoglauer.

Proceeding, Poster

Evaluation of GRB localization capabilities for the GECCO mission through simulations of the instrument BGO anti-coincidence shields. The study investigated the reconstruction of transient arrival directions from count-rate asymmetries and demonstrated localization accuracies of a few degrees on short timescales for GRB-like signals.

04/2020 – 05/2021

Gamma-Ray Burst polarization studies with AMEGO

Supervisor: Stefano Ansoldi

Co-Supervisor: Francesco Longo

Master Thesis

Study of gamma-ray polarization signatures from GRBs using simulations for the AMEGO mission. Development of an improved instrument modulation model for polarization measurements in the transition regime between Compton scattering and pair production, with applications to the discrimination of competing GRB emission models.

02/2017 – 07/2017

Estimation of the contribution of AGN host galaxies to Reionization

Supervisor: Fabio Fontanot
Co-Supervisor: Pierluigi Monaco
Bachelor Thesis
Study of the contribution of AGN host galaxies to the ionizing photon budget during the Epoch of Reionization, combining AGN luminosity functions, black hole-galaxy scaling relations, and star-forming galaxy models.

RESPONSIBILITIES

01/01/2026 – CURRENT

Joint Burst Advocate for the MAGIC & CTAO-LST1 Collaborations

La Palma, Canary Islands

15/03/2024 – CURRENT

Member of the Board of Directors for the Astronomical Observatory "Helmut Ullric"

Società Astronomica Cortina, Col Drusciè, Cortina d'Ampezzo (Belluno), Italy

20/06/2024 – 31/12/2025

Gravitational Wave Expert on Call for the MAGIC Collaboration

La Palma, Canary Islands

15/01/2024 – 31/12/2025

Burst Advocate for the CTAO-LST1 Collaboration

La Palma, Canary Islands

15/01/2024 – 31/12/2025

Burst Advocate for the MAGIC Collaboration

La Palma, Canary Islands

13/01/2025 – 12/02/2025

Data taking shift on MAGIC

Operator
La Palma, Canary Islands

29/01/2024 – 19/02/2024

Data taking shift on CTAO-LST1

Deputy Shift Leader
La Palma, Canary Islands

04/08/2023 – 25/08/2023

Data taking shift on CTAO-LST1

Operator
La Palma, Canary Islands

REFEREEING AND REVIEWING ACTIVITIES

2026 – CURRENT

Referee for Research in Astronomy and Astrophysics (RAA)

2025 – CURRENT

Internal reviewer for the MAGIC Collaboration

2024 – CURRENT

Referee for Physical Review D (APS)

COLLABORATIONS

XX/09/2026 – CURRENT

Member of the COSI Collaboration

INAF Osservatorio Astornomico di Brera, Merate

01/02/2025 – CURRENT

Member of the research team for the Crystal Eye proposal

Evaluation of the localization capabilities of Gamma-Ray Burst (GRB) signals

15/12/2024 – CURRENT

Member & co-proposer of the COST Action CA23130: Bridging high and low energies in search of quantum gravity (BridgeQG)

Working Group 1: High-energy quantum gravity theory
Working Group 2: High-energy quantum gravity experiment

01/07/2023 – CURRENT

Member of the MAGIC Collaboration

La Palma, Canary Islands

01/11/2021 – CURRENT

Member of the CTAO-LST1 Collaboration

La Palma, Canary Islands

01/01/2020 – CURRENT

Member of the Astronomical Observatory "Helmut Ullric"

Società Astronomica Cortina, Col Drusciè, Cortina d'Ampezzo (Belluno), Italy

01/08/2022 – 13/09/2023

Member of the COST Action CA18108: QG-MM "Quantum gravity phenomenology in the multi-messenger approach"

Working Group 3: Gamma rays

20/05/2021 – 30/09/2024

Member of the research team for the Galactic Explorer with a Coded Aperture Mask Compton Telescope (GECCO).

Evaluation of the localization capabilities of Gamma-Ray Burst (GRB) signals using anti-coincidence shields (BGO)

TEACHING

2026

Internal referee for bachelor degree student internship.

"Gamma-Ray Bursts analysis using GTBurst"
University of Trieste, Trieste

2025

Co-Supervisor for bachelor degree student thesis.

"Analysis of AGN VHE emissions with the MAGIC telescopes"
University of Trieste, Trieste

2024

Co-Supervisor for master degree student thesis.

"Searching for high-z GRBs in Fermi-LAT data using ML techniques."
University of Trieste, Trieste

2023 – 2024

External referee (INFN) for master degree student internships.

5 students: MC simulations with Geant4 for medical physics
1 student: simulations of HE astrophysical signals, multiwavelength data analysis, CNN
University of Trieste, Trieste

02/10/2023 – 15/12/2023

Teaching assistant for General Physics to the Mathematics curriculum

University of Udine, Udine

27/02/2023 – 09/06/2023

Teaching assistant for Analysis II to the Mathematics curriculum

University of Udine, Udine

OUTREACH AND PUBLIC ENGAGEMENT

09/2026

Studiare Fisica a Trieste

Presentation on gamma ray astronomy for incoming student orientation
University of Trieste, Italy

02/2026

Lecture at high school

Invited lecture on astrophysics and astronomy to high school students
Padova, Italy

09/2025

German Italian Physics Exchange (GIPE)

Presentation on gamma ray astronomy
University of Trieste, Italy

2022 – 2025 **Studiare Fisica a Trieste**
Presentation on gamma ray astronomy for incoming student orientation
University of Trieste, Italy

20/05/2025 **Pint of Science**
Public talk on gamma ray astrophysics: *Il Cielo nel Pallone*
Trieste, INFN & INAF, Italy

25/02/2025 – 26/02/2025 **2025 Italian Inter-regional Astronomy Olympics, judge and jury member.**
INAF, Italy

2021 – 2023 **Involved in the organization/staff of the joint INFN-INAF stand at the Trieste NEXT fair.**
University of Trieste, Italy

24/02/2022 – 26/02/2022 **2022 Italian Inter-regional Astronomy Olympics, judge and jury member.**
INAF, Italy

05/11/2021 – 13/11/2021 **International Remote Astronomy Olympics (IRAO 2021), Invigilator**
INAF, Italy

SKILLS

LANGUAGES

Italian – mother tongue
English – C1
- IELTS band 8 (23-24/07/2021)

Spanish – B1

DIGITAL SKILLS

Programming languages
- **Python**
- Wolfram Mathematica
- Fortran (77, 90, 95)
- C
- Linux Bash
- IDL
- Arduino IDE

MEGAlib
AGI System Tool Kit (STK)
Git
SLURM

OTHER CERTIFICATIONS

BLSD - Basic Life Support and Defibrillation — Basic Life Support with a Defibrillator, 23/10/2025

BLS - Basic Life Support — Basic Life Support, 03/06/2025

Nautical license — Nautical license, 04/07/2023, n° 953520

Scuba Diving — PADI Open Water Diver, 24/08/2021, n° 2109EI8387